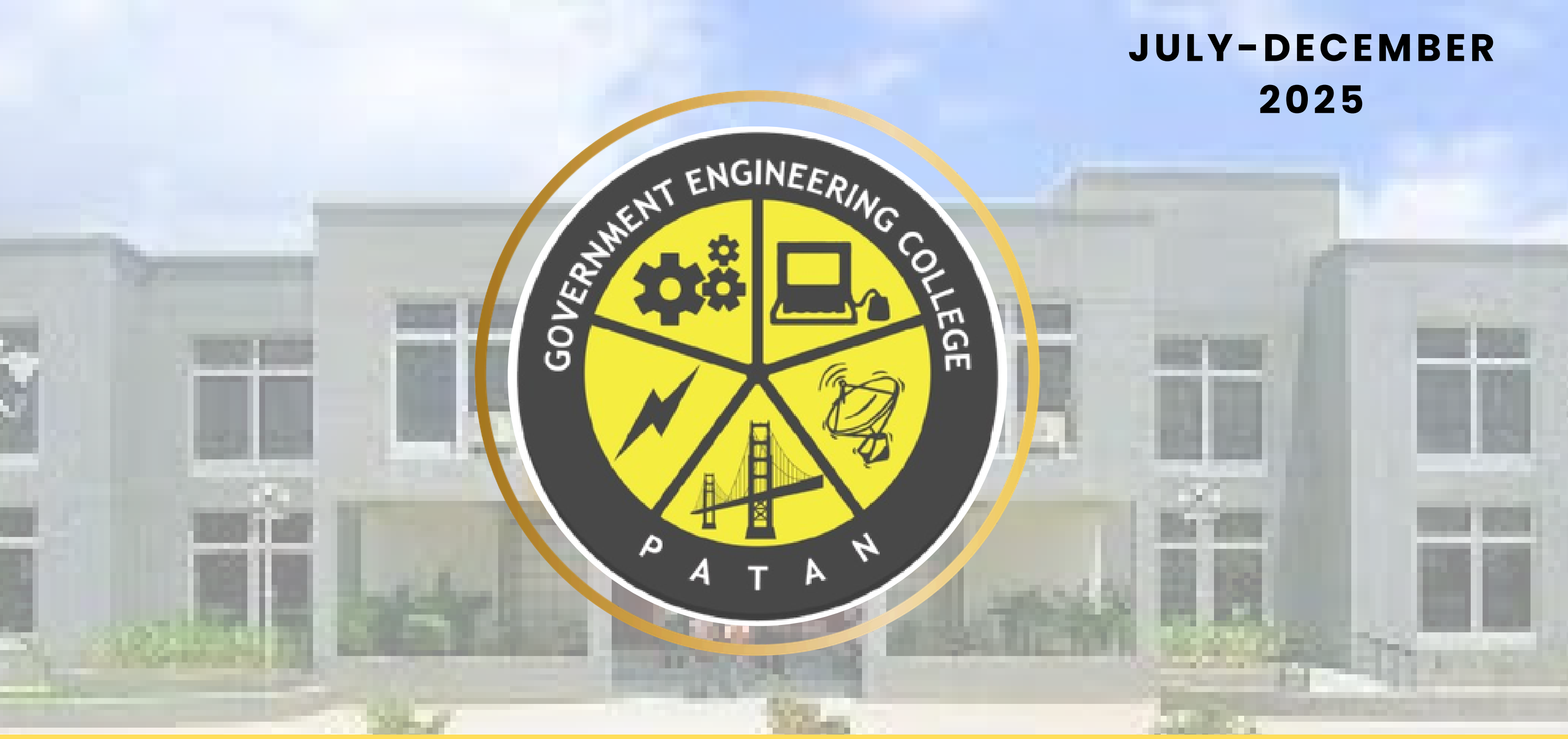

JULY-DECEMBER 2025

GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

“To prepare human resources with value-based competency for technical advancements and growth of society.”

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**

ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

"To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth."

DEPARTMENT MISSION

- **To prepare competent engineers with the ability to solve industrial problems who can raise the living standards of the society.**
- **To prepare engineers with effective communication skills, professional ethics and leadership qualities.**

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DR. DEVENDRAKUMAR H. PATEL

It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

~

Dr. Devendrakumar H. patel

HOD EC Department

Government Engineering College, Patan

ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.

Program Outcomes (POs)

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
- **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
- **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

Program Outcomes (POs)

- **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11: Life-Long Learning:** Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

DEPARTMENTAL DETAILS

ELECTRONICS AND COMMUNICATION ENGINEERING

Faculty Members

DR. DEVENDRAKUMAR H. PATEL
DESIGNATION: ASSOCIATE PROFESSOR
AREA OF INTEREST: MULTIBAND ANTENNAS, WIRELESS COMMUNICATION

DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DIGITAL COMMUNICATION,WIRELESS COMMUNICATION AND DSP

PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DSP, COMMUNICATION SYSTEMS

DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: MICROWAVE, ANTENNA, ELECTRONICS

PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: COMMUNICATION ENGG, MICROWAVE ENGG, OPTICAL FIBER COMMUNICATION

PROF.SHYAMKUMAR PANKHANIYA
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM DESIGN, VLSI TECHNOLOGY

PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: VLSI DESIGN

PROF. ROSHNIBEN CHAUDHARI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: ANTENNA, RF & MICROWAVE ENGINEERING, WIRELESS COMMUNICATION, BASIC ELECTRONICS

PROF. ANJU VASDEWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS, DIGITAL CIRCUITS AND DESIGNING, MICROPROCESSOR, MICRO CONTROLLER & INTERFACING, PROGRAMMING USING C, BASIC ELECTRONICS

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

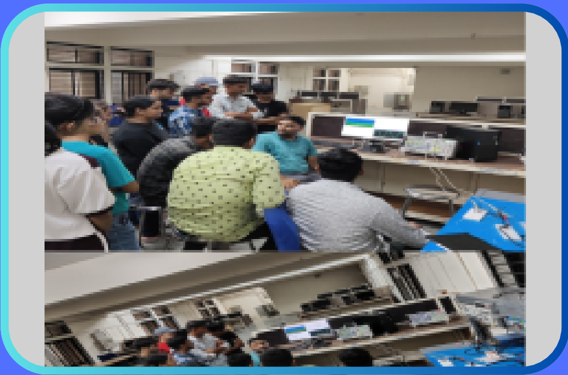
This laboratory is equipped with Analog and Digital Electronics trainers, viz.,amplifier, common emitter configuration, common collector configuration,oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR

L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.

FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench

L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.

L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing,
and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
  (ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel, 50Mhz (AFG3051C)

Scholarship Seminar (2025-26)

- A Scholarship Seminar for newly admitted 1st-year and 3rd-semester (D2D) students was organized on 10th September 2025 as per the college notice. The seminar aimed to provide comprehensive guidance on applying for various scholarship schemes such as Digital Gujarat Portal, NSP, AICTE (Saksham & Pragati), and MYSY.
- The sessions were held in the EC Seminar Room: CSE & EC Branches: 11:30 AM – 12:30 PM for MECH, CIVIL & EE Branches: 2:00 PM – 3:00 PM The seminar, conducted by Prof. M. M. Patel (Scholarship Coordinator, EC Department)

FACULTY ACTIVITIES

TRAINING and MOOC Completion

Sr. No	Sr. No	Sr. No	Place	Duration
1	Jivan Vidhya Parichay Shibir	Prof. M.B.Patel	Agrasen Foundation, Ahmedabad	28/7/2025 to 01/08/2025
2	Jivan Vidhya Parichay Shibir	Prof. M.L.Patel	Agrasen Foundation, Ahmedabad	28/7/2025 to 01/08/2025

Industrial Visit to BISAG-N

The students of the Electronics & Communication Engineering Department visited BISAG-N, Gandhinagar on 14 October 2025. The visit began with an introduction session explaining BISAG-N's objectives and national projects. 12 Students participated and gained an overview of satellite communication, GIS, and remote sensing. In the live broadcasting studio, the process of educational program recording and telecasting was demonstrated. The technical section covered satellite image processing, GIS mapping, and 3D terrain models. The use of satellite data in agriculture, disaster management, water planning, and smart cities was explained. The visit provided valuable real-world exposure and industry-level learning for students.

Poster Presentation

A Poster Presentation Activity was conducted on 25 November 2025 for Semester III ECE students at Government Engineering College, Patan. The activity was organized as a part of the Network Theory course under continuous assessment. Students prepared posters individually or in small groups on important syllabus-based topics. Key topics included Network Theorems, Two-Port Parameters, RL/RC/RLC Network Analysis, and Graph Theory. Faculty members interacted, evaluated, and assessed students using a predefined rubric. Students explained concepts using diagrams, equations, practical applications, and answered questions confidently. The activity improved conceptual understanding, technical knowledge, presentation skills, and outcome-based learning.

Project Presentation

Date: 25th November 2025
Time: 11:00 A.M. onwards
Venue: Electronics and Communication Engineering Department (L-203)
Participants: 38 students (Second Year EC)

The Project Presentation, organized by the Electronics and Communication Engineering Department as part of academic activities for second-year EC students, aimed to strengthen practical understanding, innovation, teamwork, and technical presentation skills by bridging theoretical knowledge with real-world applications. Conducted under the guidance of Prof. R. A. Chaudhari with evaluation by Dr. H. L. Judal, Prof. M. L. Patel, and Prof. Viral Pansiniya, the event featured group presentations of mini projects through hardware models, breadboard implementations, and software demonstrations. Students explained design concepts and applications while interacting with the jury, receiving constructive feedback. Overall, the activity enhanced confidence, problem-solving ability, and hands-on learning, contributing effectively to Outcome-Based Education (OBE) and aligning with NBA Criterion 2 – Teaching–Learning Processes.

Inaugural Function of VLSI Club

The Department of Electronics and Communication Engineering, Government Engineering College, Patan, successfully organized the VLSI Club Opening Ceremony on 15 October 2025 at the Seminar Room, inaugurating the club under the guidance of the Principal, senior faculty, and department leadership. The event aimed to enhance student awareness of VLSI and semiconductor technologies, promote industry-oriented and hands-on learning, and bridge the gap between academics and industry requirements. Faculty members highlighted the role of VLSI in modern electronics, AI hardware, IoT, and semiconductor design, while outlining future activities such as workshops, expert talks, and practical training. Overall, the event motivated ECE students towards industry-relevant skills and strongly supported outcome-based education in line with NBA and NAAC standards.

PROGRAMS PARTICIPATE BY FACULTIES

Sr. No	Title	Name of the faculty	Dept.	Place	Date	Duration
1	AICTE QIP PG Certificate Programme on Semiconductor Technologies: Design, Fabrication and Applications under the AICTE QIP Sponsored Category scheduled from June to December-2025	Prof. D. H. Patel	EC	Offline (IIIT Surat) + Online	16/06/2025 to 21/12/2025	June-2025 to December-2025
2	Teaching Tools for Next Generation Students	Prof. M. B. Patel	EC	Online	4/12/2025	01 Day
3	Teaching Tools for Next Generation Students	Prof. R. A. Chaudhary	EC	Online	4/12/2025	01 Day
4	From Vision to Velocity: Workforce Development for the Semiconductor Leap	Prof.M.L.Patel	EC	Ganpat University	9/10/2025	01 Day

Webinar on Digital Marketing & E-Commerce

The NSS Unit of Government Engineering College, Patan, in collaboration with the Gymkhana Cell, organized a seminar and activities under Vikas Saptah on 7th October 2025. The event focused on creating awareness about E-Commerce and digital entrepreneurship among students. An expert webinar and an online quiz competition were conducted, with active participation from students and faculty. The program concluded with the Vikas Saptah pledge, encouraging students to contribute to national development.

Manushya Gaurav Din

On the occasion of Manushya Gaurav Din, an inspiring program was organized with the presence of respected professors and guests. The event focused on human values, self-respect, positive thinking, and equality. Thought-provoking talks and spiritual reflections encouraged respect for oneself and others, leaving participants motivated and inspired.

Yuva Apda Mitra Camp

The 7-day Yuva Apda Mitra Camp was successfully conducted at SRPF Camp, Madana (Banaskantha), offering an enriching blend of knowledge, discipline, and hands-on experience in disaster management. From the very first day, the camp set a strong foundation by introducing participants to the importance of disaster preparedness and the critical role of trained volunteers in saving lives.

Throughout the camp, participants received intensive training in basic first aid, Golden Hour response, bleeding control, CPR, patient assessment, and safe lifting and transportation techniques. Practical demonstrations and repeated drills helped build confidence and clarity in handling real emergency situations.

The training further expanded to fire safety, water rescue, swimming techniques, knot-tying, cyclone awareness, and flood-rescue methods, making the learning holistic and field-oriented. A special visit to Dantiwada Dam provided real-time exposure to water rescue operations, enhancing practical understanding.

Physical training sessions improved stamina and discipline, while motivational talks inspired volunteers to serve society with responsibility and courage. Overall, the camp proved to be a powerful learning experience that transformed participants into more confident, skilled, and socially responsible disaster response volunteers.

Drawing Competition

The NSS Unit of GEC Patan organized a Drawing Competition to pay tribute to Bhagwan Birsa Munda on the occasion of 150 Years of Janjatiya Gaurav Varsh. The event was held on 10th November with great enthusiasm under the theme "The Life and Legacy of Bhagwan Birsa Munda."

Essay Writing Competition.

The Essay Writing Competition was organized under Janjatiya Gaurav Varsh on the theme "Bhagwan Birsa Munda's Life and Contribution." The event aimed to encourage students to explore and express the inspiring journey, values, and sacrifices of Bhagwan Birs Munda, one of India's greatest tribal leaders. The event was conducted on 11th November, 2025, from 3:00 p.m. to 4:00 p.m. at the Seminar Hall, EC Department.

National Adventure Camp

NSS Volunteer Mal Jyoti, a student of 5th Semester CSE, was selected to represent the institution at the National Adventure Camp held at Dharamshala, Himachal Pradesh, from 30th November to 9th December 2025. Participation in this prestigious national-level camp provided her with valuable exposure to adventure activities, leadership training, teamwork, and self-discipline, while fostering resilience and confidence. Her selection and active involvement reflect her dedication to the ideals of the National Service Scheme (NSS) and bring pride to the college.

Thalassemia Minor Screening Camp

The NSS Unit of Government Engineering College Patan, in collaboration with the Indian Red Cross Society, organized a Thalassemia Minor Screening Camp at the Mechanical Workshop. The camp aimed to promote early detection and awareness about Thalassemia. A medical team, with the support of NSS volunteers, conducted smooth sample collection and guided students about preventive healthcare. Around 360 students participated, making the initiative successful in spreading health awareness and social responsibility.

Mental Health Awareness Seminar

The Mental Health Awareness Seminar was organized on 16th December 2025 by NSS GEC Patan in collaboration with the Psychology Cell of Government Engineering College Patan under the Mind Strong Gujarat initiative. The seminar aimed to promote mental well-being and emotional awareness among students. The session was delivered by Mr. Vidit Sharma, Youth Ambassador, Government of Gujarat, who spoke on stress management, self-awareness, emotional resilience, and the importance of seeking help. Students actively participated, making the session interactive and impactful. The seminar concluded with a strong message promoting positive life choices and mental strength.

World Meditation Day

The NSS Unit of Government Engineering College, Patan celebrated World Meditation Day on 21st December 2025 in front of the Centre of Excellence, with enthusiastic participation from various faculty members and students. The programme featured the collective performance of Suryanamaskar along with a series of yoga and meditation exercises, highlighting the importance of physical fitness, mental well-being, inner peace, self-discipline, and the adoption of a healthy and balanced lifestyle.

Know Your Leader Program

On 25 December 2025, **Abhishek Hirenbhai Mehta,** an NSS volunteer and student of Government Engineering College, Patan (GTU), represented Gujarat as a speaker at the prestigious “Know Your Leaders” programme held at the Constitution Hall of Parliament House, New Delhi, commemorating the birth anniversaries of Pandit Madan Mohan Malaviya and Atal Bihari Vajpayee. Selected through a rigorous UGC-led national process, he delivered a powerful speech in Gujarati highlighting Malaviyaji’s life and ideals, showcasing Gujarati culture and patriotism; the event was organized by PRIDE, Lok Sabha Secretariat, led by Lok Sabha Speaker Shri Om Birla, and was broadcast live on Sansad TV, marking a proud moment for Gujarat.

EDITORIAL TEAM

Patron

Dr. D. H. Patel
HOD, Electronics and Communication Department

Editors-in-Chief

Dr. M. B. Patel - Electronics & Communication Engineering

Content & Design Student Team

Joshi Parv Kashyap (240220111004 - EC)
Verma Samir Santoshbhai (240220111033 - EC)
Dahivat Ahir (250022011101 - EC)
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